

Concept 2 | Differentiation and Tangent Lines

English Student Edition • Focused formulas and guided practice

Formula opener

Gradient

$$m = f'(x_1)$$

Tangent

$$y - y_1 = m(x - x_1)$$

Normal

$$m_n m = -1$$

Worked Solutions

Differentiation and Tangent Lines

Q001 Written

Differentiate the four Warm Up demands: $y = x^2 - 3x + 2$, $y = (2x + 1)^3$, $V = \pi r^2 h$ with respect to (r), and find $f'(-1)$ for $f(x) = x^3 - 3x + 1$.

Solution

1. Use $(x^n)' = nx^{n-1}$. For a composite cubic, expand or use the chain rule. Treat (h) as constant in dV/dr , then substitute $x = -1$ into $3x^2 - 3$.

Final answer

$2x - 3; 6(2x + 1)^2; 2\pi rh; 0$

Q002 Written

Differentiate the six printed Try functions, the two variable-labelled functions $S = \pi r^2$ and $h = vt - \frac{1}{2}gt^2$, and evaluate $f'(2)$, $f'(-3)$ for $f(x) = 4x^3 - 3x^2 + 2$.

Solution

1. Differentiate each term, keeping non-differentiated variables constant. For the values, first form $f'(x) = 12x^2 - 6x$, then substitute.

Final answer

$3, -6x + 8, 2x + 3, 6x^2 - 8x + 6, 6(2x - 3)^2, 4x^2 + x - 1; 2\pi r, v - gt; 36, 126$

Worked Solutions

Differentiation and Tangent Lines

Q003 **Written**

Differentiate every source Exercise function 18 polynomial/product forms, the six functions with variables indicated in brackets, and evaluate the two values in each of Exercises 3–5.

Solution

1. Bring down the exponent and subtract one. Expand products before differentiating when no chain rule is being used; then substitute the stated input.

Final answer

Apply the power rule term-by-term; for a product or power, expand first. The value sets are: Exercise 3 (33,65); Exercise 4 $-24, -3$; Exercise 5 (3,28).

Worked Solutions

Differentiation and Tangent Lines

Q004 Written

For $f(x) = \sqrt[3]{x^5} + \frac{1}{x^2}$, find $f'(1) + f'(-1)$.

Solution

1. Write $f(x) = x^{5/3} + x^{-2}$, so $f'(x) = \frac{5}{3}x^{2/3} - 2x^{-3}$. Hence $f'(1) = -\frac{1}{3}$, $f'(-1) = \frac{11}{3}$, and the sum is $\frac{10}{3}$.

Final answer

$$\frac{10}{3}$$

Q005 Written

Find the quadratic $f(x) = ax^2 + bx + c$ satisfying $f'(0) = -2$, $f'(2) = 2$, $f(1) = 2$.

Solution

1. Since $f'(x) = 2ax + b$, $b = -2$, $4a + b = 2$, and $a + b + c = 2$. Thus $a = 1$, $b = -2$, $c = 3$.

Final answer

$$f(x) = x^2 - 2x + 3$$

Worked Solutions

Differentiation and Tangent Lines

Q006 Written

Find the quadratic satisfying $f'(0) = 2$, $f'(1) = 4$, $f(2) = 6$.

Solution

1. From $2ax + b$: $b = 2$, $2a + b = 4$, and $4a + 2b + c = 6$, giving $a = 1$, $b = 2$, $c = 0$.

Final answer

$$f(x) = x^2 + 2x$$

Q007 Written

Find the four quadratics satisfying the four source triples of derivative and function conditions.

Solution

1. For each item set $f(x) = ax^2 + bx + c$, use $f'(x) = 2ax + b$, substitute the two derivative conditions and one function condition, then solve the three linear equations.

Final answer

$$x^2 - 4x + 8; -2x^2 + 3x + 3; -5x^2 + 26x - 14; -\frac{3}{2}x^2 + 2x + \frac{3}{2}$$

Worked Solutions

Differentiation and Tangent Lines

Q008 Written

Find the quadratic ($f(x)$) satisfying $f'(2) + f'(4) = 0$, $f(3) = 0$, $f(1) = 12$.**Solution**

1. With $f'(x) = 2ax + b$, $12a + 2b = 0$, $9a + 3b + c = 0$, and $a + b + c = 12$. Solving gives $a = 3$, $b = -18$, $c = 27$.

Final answer

$$f(x) = 3x^2 - 18x + 27$$

Q009 Written

Find the tangent line to $y = x^2 - 4x + 1$ at $(3, -2)$.**Solution**

1. The gradient is $f'(3) = 2$. Use $y + 2 = 2(x - 3)$, hence $y = 2x - 8$.

Final answer

$$y = 2x - 8$$

Worked Solutions

Differentiation and Tangent Lines

Q010 Written

Find the tangent line for the three source functions at their stated points: $y = x^2 + 3$ at $(-2, 7)$, $y = x^2 - 3x + 2$ at $((1, 0))$, and $y = -x^3 + 2x + 4$ at $(-2, 8)$.

Solution

1. Differentiate, evaluate the gradient at the given x-coordinate, and use $y - y_1 = m(x - x_1)$ for each point.

Final answer

$$y = -4x - 1; y = -x + 1; y = -10x - 12$$

Q011 Written

Find the tangent line for all nine source Exercise functions at their given points.

Solution

1. For each item compute $m = f'(a)$, insert $((a, f(a)))$ into $y - f(a) = m(x - a)$, then rearrange.

Final answer

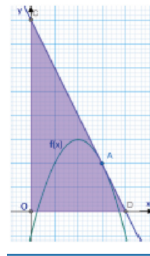
$$y = 4x - 5; y = 4x - 1; y = x - 4; y = x - 7; y = -4x + 3; y = 7x - 3; y = 9x - 16; y = -x - 2; y = x + 1$$

Worked Solutions

Differentiation and Tangent Lines

Q012 Written

For $f(x) = -x^2 + 4x - 1$, the tangent at x-coordinate 3 meets the axes at (D,C). Find the area of triangle (ODC).



Solution

- The tangent is $y = -2x + 8$, so $D = (4, 0)$ and $C = (0, 8)$. Therefore area $= \frac{1}{2}(4)(8) = 16$. The ministry graph is retained beside the demand.

Final answer

16 square units

Worked Solutions

Differentiation and Tangent Lines

Q013 Written

Solve the two source Warm Up demands: tangents from $C(0,2)$ to $y = x^2 - 3x + 6$, and tangents to $y = x^3 + 4x^2 - 3$ with gradient 3.

Solution

1. Let the point of tangency be $((a, f(a)))$. Equate the tangent line through the external point, or set $(f'(a))$ equal to the required gradient, then solve for (a) .

Final answer

$$y = -7x + 2, y = x + 2; \quad y = 3x + 15, y = 3x - \frac{95}{27}$$

Q014 Written

Solve the two source Try demands: tangents from $C(1, -1)$ to $y = x^2 - x + 3$, and tangents to $y = x^3 + 2x - 3$ with gradient 5.

Solution

1. Use $((a, f(a)))$ and $m = f'(a)$. The first condition gives $a = 3, -1$; the second gives $a = 1, -1$. Substitute each point into the tangent formula.

Final answer

$$y = 5x - 6, y = -3x + 2; \quad y = 5x - 5, y = 5x - 1$$

Worked Solutions

Differentiation and Tangent Lines

Q015 Written

Solve all six source Exercise demands involving external points or prescribed gradients.**Solution**

1. Let the contact point be $((a, f(a)))$, form $y - f(a) = f'(a)(x - a)$, and impose the external point or gradient condition. Retain every real value of (a) .

Final answer

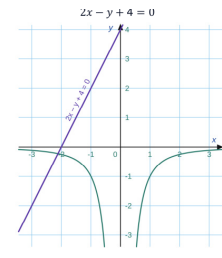
$$y = 6x - 8, y = -2x; y = x + 1, y = -7x + 25; y = -5x + 6, y = 3x - 2; y = 9x - 11, y = 9x + 21; y = -x; y = -2x + 16, y = -2x - 8$$

Worked Solutions

Differentiation and Tangent Lines

Q016 Written

For $f(x) = -1/x^2$, find the tangent line parallel to $2x - y + 4 = 0$.



Solution

1. The given line has gradient 2. Since $f'(x) = 2/x^3$, $2/a^3 = 2$ gives $a = 1$. The tangent at $(1, -1)$ is $y + 1 = 2(x - 1)$, so $y = 2x - 3$. The ministry graph is retained beside the demand.

Final answer

$$y = 2x - 3$$

Worked Solutions

Differentiation and Tangent Lines

Q017 MCQ

The derivative of x^n is:**Solution**

1. Bring down n and subtract 1 from the exponent.

Correct option: B**Final answer**

$$nx^{n-1}$$

Q018 MCQ

The derivative of $V = \pi r^2 h$ with respect to r is:**Solution**

1. Treat h as a constant.

Correct option: B**Final answer**

$$2\pi rh$$

Worked Solutions

Differentiation and Tangent Lines

Q019 MCQ

For $f(x) = x^3 - 3x + 1$, $f'(-1)$ is:

Solution

1. $f'(x) = 3x^2 - 3$.

Correct option: C

Final answer

0

Q020 MCQ

The challenge function can be written as:

Solution

1. Convert the cube root and reciprocal to powers.

Correct option: A

Final answer

$$x^{5/3} + x^{-2}$$

Worked Solutions

Differentiation and Tangent Lines

Q021 MCQ

For $f(x) = ax^2 + bx + c$, $(f'(x))$ is:**Solution**

1. Differentiate the quadratic.

Correct option: B**Final answer**

$$2ax + b$$

Q022 MCQ

The condition $f'(0) = -2$ gives:**Solution**

1. Substitute $x = 0$ into $2ax + b$.

Correct option: B**Final answer**

$$b = -2$$

Worked Solutions

Differentiation and Tangent Lines

Q023 MCQ

The warm-up quadratic is:**Solution**

1. Solve the three coefficient equations.

Correct option: A

Final answer

$$x^2 - 2x + 3$$

Q024 MCQ

The challenge coefficient a is:**Solution**1. Use $f'(2) + f'(4) = 0$, then the two function values.

Correct option: C

Final answer

3

Worked Solutions

Differentiation and Tangent Lines

Q025 MCQ

The tangent gradient at $x=a$ is:**Solution**

1. The derivative gives the instantaneous gradient.

Correct option: C**Final answer**

$$f'(a)$$

Q026 MCQ

The tangent to $y = x^2 - 4x + 1$ at $(3, -2)$ is:**Solution**

1. Use gradient 2 and point $3, -2$.

Correct option: B**Final answer**

$$y = 2x - 8$$

Worked Solutions

Differentiation and Tangent Lines

Q027 MCQ

In the triangle challenge, the x-intercept of the tangent is:**Solution**1. The tangent is $y = -2x + 8$.

Correct option: C

Final answer

4

Q028 MCQ

The area of triangle (ODC) in the challenge is:**Solution**

1. Use one half base times height.

Correct option: C

Final answer

16

Worked Solutions

Differentiation and Tangent Lines

Q029 MCQ

When the contact point is not given, set it to:**Solution**

1. Use an unknown contact x-coordinate.

Correct option: B**Final answer** $(a, f(a))$

Q030 MCQ

The gradient of $2x - y + 4 = 0$ is:**Solution**

1. Rearrange to $y = 2x + 4$.

Correct option: D**Final answer**

2

Worked Solutions

Differentiation and Tangent Lines

Q031 MCQ

For $f(x) = -1/x^2$, $(f'(x))$ is:**Solution**1. Write $f = -x^{-2}$ and differentiate.

Correct option: B

Final answer

$$2/x^3$$

Q032 MCQ

The parallel tangent in the challenge is:**Solution**1. The contact point is $a = 1$.

Correct option: B

Final answer

$$y = 2x - 3$$

Worked Solutions

Differentiation and Tangent Lines

Quiz WRITTEN 1 Concept Quiz · Written**Differentiate** $3x^3 - 2x$.**Solution**

1. Apply the power rule.

Final answer

$$9x^2 - 2$$

Quiz WRITTEN 2 Concept Quiz · Written**Find the quadratic with** $f'(0) = 1, f'(1) = 3, f(1) = 2$.**Solution**

1. Use $f'(x) = 2ax + b$ and solve for (a,b,c).

Final answer

$$x^2 + x$$

Worked Solutions

Differentiation and Tangent Lines

Quiz WRITTEN 3 Concept Quiz · Written

Find the tangent to $y = x^2$ at $((2,4))$.

Solution

1. Gradient $f'(2) = 4$, then use point-slope form.

Final answer

$$y = 4x - 4$$

Quiz WRITTEN 4 Concept Quiz · Written

For $f(x) = -1/x$, find the tangent parallel to $y = x + 2$.

Solution

1. Set $f'(a) = 1$, giving $a = 1$.

Final answer

$$y = x - 4$$

Worked Solutions

Differentiation and Tangent Lines

Quiz MCQ 5 **Concept Quiz · MCQ**

The derivative of a constant is:

Solution

1. A constant has zero rate of change.

Correct option: A

Final answer

0

Quiz MCQ 6 **Concept Quiz · MCQ**

For $f(x) = ax^2 + bx + c$, the coefficient equations come from:

Solution

1. There are three unknown coefficients.

Correct option: B

Final answer

two derivatives and one function value

Worked Solutions

Differentiation and Tangent Lines

Quiz MCQ 7 [Concept Quiz · MCQ](#)

The tangent equation in point-slope form is:

Solution

1. Use the given point and tangent gradient.

Correct option: A

Final answer

$$y - y_1 = mx - x_1$$

Quiz MCQ 8 [Concept Quiz · MCQ](#)

A tangent parallel to $y = 3x - 1$ has gradient:

Solution

1. Parallel lines have equal gradients.

Correct option: C

Final answer

3

Answer key

Use the question number shown on each page.

Q001 $2x - 3; 6(2x + 1)^2; 2\pi rh; 0$

Q008 $f(x) = 3x^2 - 18x + 27$

Q019 C

Q002 $3, -6x + 8, 2x + 3, 6x^2 - 8x + 6, 6(2x - 3)^2, 4x^2 + x - 1; 2\pi r, v - gt; 36, 20$

Q009 $y = 2x - 8$

Q020 A

Q003 Apply the power rule term-by-term; for a product or power, expand first. The value sets are: Exercise 3 (33,65); Exercise 4 $-24, -3$; Exercise 5 (3,28).

Q010 $y = -4x - 1; y = -x + 1; y = -10x - 12$

Q021 B

Q011 $y = 4x - 5; y = 4x - 1; y = x - 4; y = x - 7; y = -4x + 3; y = 7x - 3; y = 9x - 16; y = -x - 2; y = -4x - 1$

Q022 B

Q012 16 square units

Q023 A

Q013 $y = -7x + 2, y = x + 2; y = 3x + 15, y = 3x - \frac{85}{27}$

Q024 C

Q014 $y = 5x - 6, y = -3x + 2; y = 5x - 5, y = 5x - 1$

Q025 C

Q004 $\frac{10}{3}$

Q015 $y = 6x - 8, y = -2x; y = x + 1, y = -7x + 25; y = -5x + 6, y = 3x - 2; y = 9x - 11, y = 9x + 21; y = -x; y = -2x + 16, y = -2x - 4$

Q026 B

Q005 $f(x) = x^2 - 2x + 3$

Q016 $y = 2x - 3$

Q027 C

Q006 $f(x) = x^2 + 2x$

Q017 B

Q028 C

Q007 $x^2 - 4x + 8; -2x^2 + 3x + 3; -5x^2 + 26x - 14; -\frac{3}{2}x^2 + 2x - \frac{3}{2}$

Q018 B

Q029 B

Q030 D

Q031 B

Q032 B

Quiz WRITTEN 1 $9x^2 - 2$

Quiz WRITTEN 2 $x^2 + x$

Quiz WRITTEN 3 $y = 4x - 4$

Quiz WRITTEN 4 $y = x - 4$

Quiz MCQ 5 A

Quiz MCQ 6 B

Quiz MCQ 7 A

Quiz MCQ 8 C